

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To recommend an item to the user using content-based filtering approach	
Experiment No: 11	Date:	Enrolment No: 92301733023

Aim: To recommend an item to the user using content-based filtering approach

IDE: Google Colab **Theory:**

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

Content-Based Recommendation System: Content-Based systems recommends items to the customer similar to previously high-rated items by the customer. It uses the features and properties of the item. From these properties, it can calculate the similarity between the items.

In a content-based recommendation system, first, we need to create a profile for each item, which represents the properties of those items. From the user profiles are inferred for a particular user. We use these user profiles to recommend the items to the users from the catalog.

Item profile:

In a content-based recommendation system, we need to build a profile for each item, which contains the important properties of each item. For Example, If the movie is an item, then its actors, director, release year, and genre are its important properties, and for the document, the important property is the type of content and set of important words in it.

Let's have a look at how to create an item profile. First, we need to perform the TF-IDF vectorizer, here TF (term frequency) of a word is the number of times it appears in a document and The IDF (inverse document frequency) of a word is the measure of how significant that term is in the whole corpus.

User profile:

The user profile is a vector that describes the user preference. During the creation of the user's profile, we use a utility matrix that describes the relationship between user and item. From this information, the best estimate we can decide which item the user likes, is some aggregation of the profiles of those items.

Advantages and Disadvantages:

- **Advantages:**
- No need for data on other users when applying to similar users.
- Able to recommend to users with unique tastes.
- Able to recommend new & popular items

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- Explanations for recommended items.
- **Disadvantages:**
- Finding the appropriate feature is hard.
- Doesn't recommend items outside the user profile.

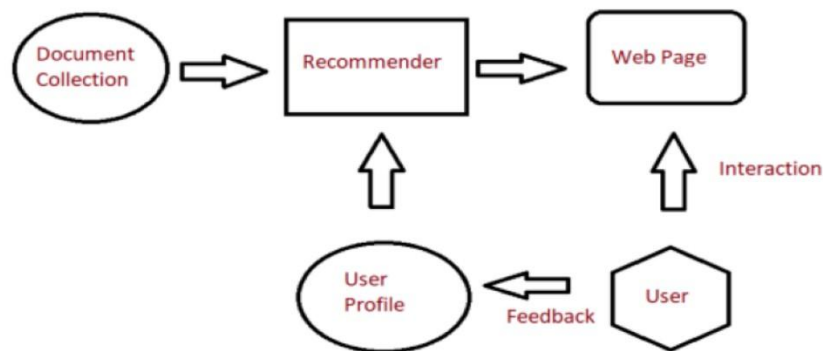


Fig: Recommender System

Pre Lab Exercise:

1. When can you use content-based recommendation system approach?

When we want to recommend items based on a user's past interest or item features like genre, keywords, etc. It works well when we don't have data of other users.

2. Write advantages of content-based filtering approach

- No need of other users' data
- Works for unique user preferences
- Can recommend new items
- Easy to understand (based on features)

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3. Write disadvantages of content-based filtering approach

- Hard to choose correct features
- Gives only similar type of items
- Lacks variety in recommendations

Program (Code):

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
df=pd.read_csv("/content/movies.csv")
#preprocessing step
df["genres"]=df["genres"].fillna("")
df["overview"]=df["overview"].fillna("")
df["content"]=df["genres"]+" "+df['overview']
#TF-IDF Vectorizer
tfidf=TfidfVectorizer(stop_words="english")
tfidf_matrix=tfidf.fit_transform(df["content"])
cosine_sim=cosine_similarity(tfidf_matrix,tfidf_matrix)
#dropping the duplicates
indices=pd.Series(df.index,index=df["original_title"]).drop_duplicates()
def recommendation_movie(title,topn=10):
    if title not in indices:
        return "Movie is not found!"
    idx=indices[title]
    sim_scores=list(enumerate(cosine_sim[idx]))
    sim_scores=sorted(sim_scores,key=lambda x:x[1],reverse=True)
    sim_scores=sim_scores[1:topn+1]
    movie_indices=[i[0] for i in sim_scores]
    return df["original_title"].iloc[movie_indices]
recommendation_movie("Inception")
```

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Results:

original_title	
2897	Cypher
4401	The Helix... Loaded
134	Mission: Impossible - Rogue Nation
1786	Flatliners
2389	Renaissance
1930	Stone Cold
914	Central Intelligence
322	The Fifth Element
1271	Pandorum
1267	Duplex

dtype: object

Observation:

The system recommends movies based on similarity using TF-IDF and cosine similarity. Results are accurate but limited to similar types of movies.

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Post Lab Exercise:

1. **What changes you feel can be done, to improve the system that you have designed, as a part of in-house lab exercise.**

- Add more features like cast, director
- Show top 10 results instead of 5
- Improve UI (posters, ratings)
- Use hybrid recommendation (content + collaborative)

2. **Recommend top-10 “series” using IMDB movie dataset**

(Link: <https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000movies-and-tv-shows>). Attach the code with output. Also, write your observation for the same.